The Bicylinder: Steinmetz Solid With a 2 Way Restricted Rolling Path.

By adenda2 in Workshop > Woodworking 6,840 15 15 Featured Winner

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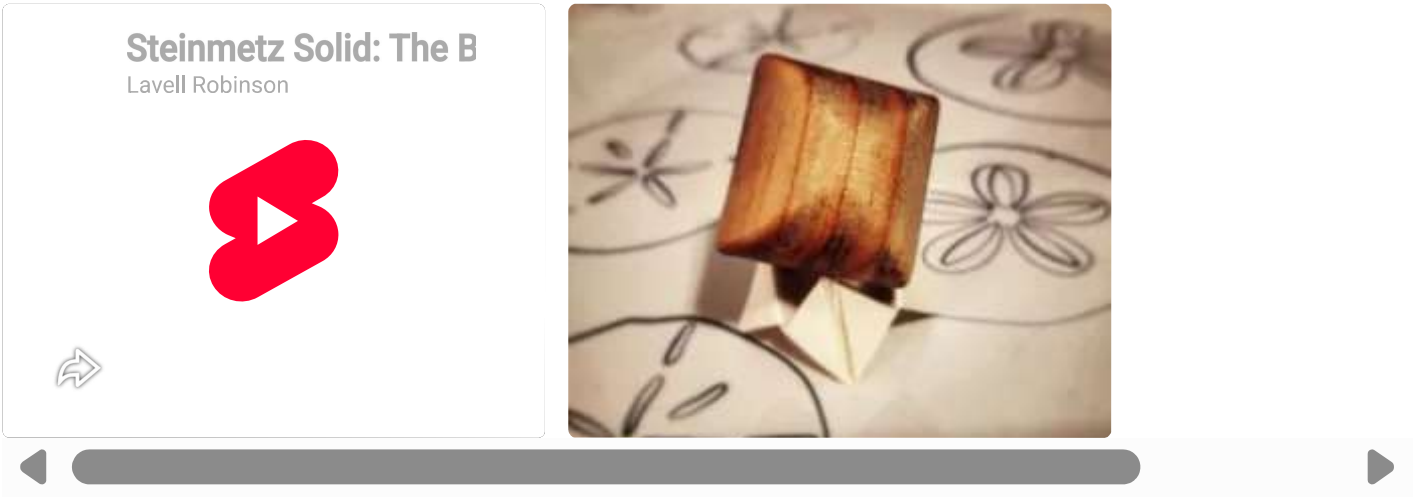

Here lies a Steinmetz solid: a physical model of two cylinders intersecting at right angles to each other on the same plane. From the Z axis it looks like a square. From x and y axis, it looks like a circle. As such, it can roll in only 2 directions.

Supplies

Wood

Lathe

Lathe tools

Sand paper

Patience

Step 1: Background: Steinmetz

Charles Steinmetz is known most for the Steinmetz's equation,

>>>>sometimes called the power equation, which is an empirical equation used to calculate the total power loss per unit volume in magnetic materials when subjected to external sinusoidally varying magnetic flux.<<<<
>>>>https://en.m.wikipedia.org/wiki/Steinmetz%27s_equation

Charles Steinmetz was a mathematician and an electrical engineer, as well as a professor at Union College.

His work helped bring forth AC current, electric motors, metal halide lamp, and a few other things.

The Steinmetz Solid:
There are a few shapes, for which there are varying mathematical equations and ways to solve for the area, which belong to the steinmetz solids: these specific shapes comprise the essence of the intersection between intersecting cylinders. This one is called a Bicylinder.

For more information on steinmetz solids, go to wolfram mathworld at...

http://mathworld.wolfram.com/SteinmetzSolid.html

Step 2: Why Wood? and Why This?

I've always been intrigued by tops, blocks, and things roll, spin, stack, and the like...

I can still remember my grandfather bringing me out to his shop when I was 7, and cutting out 2 tops for my brother and I.

While researching a few things I might print with my new 3d printer, I ran across this little wonder called the bicylinder. It reminded me of the amazing solids of

constant width, the kururin, as well as the sphericons, and the oloid. And if you haven't heard of those, you're in for a treat.

As luck would have it, my prints never stuck quite right, and then the Z axis stripped and the printer broke.. (returned the mp mini, currently awaiting the elegoo mars.. so excited)

Well, the 2 days wait time for the new printer seemed too long to try out this nice little intrigue, so I had to make it on my own.

Is just so happens, that to my knowledge, there is not one person who has made this shape out of wood, or on a lathe... and that's for a good reason: It's not easy. But with some eginuity, patience, and overall macgyver-ing, it turned out pretty good. Pun intended. ;)

Step 3: The Process:

With my setup, I have no jaw chuck to hold the piece.... my lathe is a cheaper lathe from harbor freight, which I acquired roughly 8 years ago.

I started with a piece of square about 2x2, put it on the stock turning chuck, with the live center on the tail stock, and turned it into a dowel.

Now, this isn't too complicated in theory; turn a dowel, cut it, and turn it at 90 degrees. The fun comes in centering a dowel sideways, with nothing to hold it with.

To hold this piece, I turned another dowel, which is hollowed out to fit flush over .y live center rail stock. The head just has a face plate with a 2x4 turned flat. In essence I butted it up between 2 flat surfaces, with only the pressure from the tail stock, and friction to hold it in place.

So turn a dowel, measure the diameter, and then cut a length just slightly longer than that to turn with.

Now that the piece is turned, cut, rotated, a d mounted tight, you ca. Start tu turn again, a simple straight cylinder... only this time it will complete the shape.

As I went I had to tap, tap, tap with a hammer to center the piece as I slowly carved lines into the center. Tou should be able to just BARELY get a full line in the center of it... if you go too far, or are not centered, the diameters of the intersecting cylinders will not match, and it wont roll both ways.

After the lathe, I pulled it off and sanded it down to 250, put some wax on, and gave it a roll.

Let me know in the comments if you make, this, lathe it, 3d print it, etc. also, ideas on centering it better with a home-made lathe chuck I just didn't take time to make.